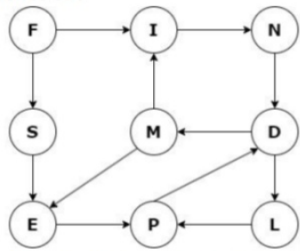


1. Sherlock Holmes found himself in a dire situation where James Moriarty imprisoned his assistant Watson in a maze. The maze is made of 9 different rooms. Each room has a part of the key with an alphabet written on it. Moriarty tells Sherlock that only if he can attach all the key chain parts in exactly this order: “F-I-N-D-M-E-P-L-S”, then he can unlock the room with that key where Watson is kept. The map of the maze is given below:



a) [CO1] Now you know something about a graph traversal algorithm that travels from the source to as far as possible along each branch (depth-wise traversal). Simulate/Draw the traversal and fill up the following table (in your script) with the start and ending time for each node you achieved while doing this task. Keep in mind, wherever you have to choose between different neighbors to explore, follow the sequence that will make the key chain parts in the sequence that Sherlock needs.

Node	D	E	F	I	M	S	L	P	N
Start									
End									

b) [CO1] When Sherlock tries to save his friend, water suddenly fills the rooms. He needs to find a way to contain the water somehow so that the water doesn't overflow from one place to another. So, he created different isolated closed groups of rooms where he can travel anywhere in that particular group, meaning they have a path from any room to any other room inside the group. After doing so, he finds out that he can make a few of these groups of rooms inside this maze. A group can contain 1 to N number of rooms where N is the total number of vertices, and no two groups can share the same room.

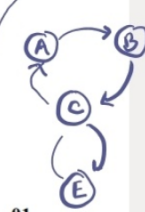
- i) **Mention** the name of a suitable algorithm for this task.
- ii) **Show** simulation for the graph given above.

You can reuse/refer to your answer from question (a) to reduce work.

06

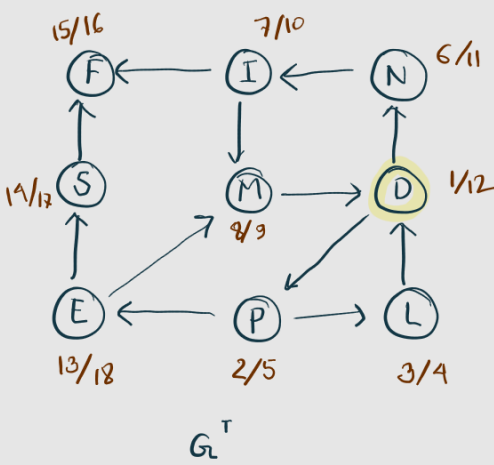
Group এর সব path এ যাওয়া যায়

→ SCC



01
03

(b) SCC হতে কারণ any room to any room inside group যাওয়া যায়।



{D, P, L, M, N, I}

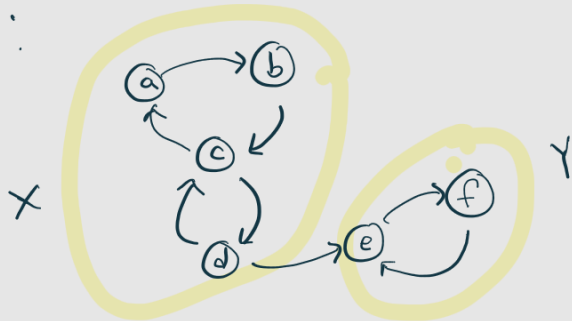
→ SCC হতে কারণ
D থেকে শুরু হয়ে
D তে ফিরবে

{E, S, F} → SCC না কারণ E তে
ফিরে না

Question: If we consider SCC as a single node will there be any cycle?

Ans: No

Example :



* SCC সমূহকে একটা node বানায় দিলে কোন cycleই থাকবে না



* DAG (Directed Acyclic Graph) হয়ে যাবে
→ cycle নাই

1

You have just joined a new university and are exploring the campus, which consists of several buildings connected by one-way bridges. These bridges form a directed graph, where each node represents a building and each edge represents a bridge between two buildings. You aim to analyze how the buildings and bridges are connected to understand the campus layout better and improve your navigation between classes and facilities.

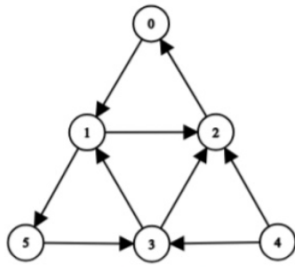


Figure 1

[Fall 24]

CO2 a) **Mention** which graph representation will let you **check** whether any two buildings have a bridge between them in $O(1)$ time. **Adjacency matrix**

01

b) From Figure 1

CO1 i. **Simulate** an algorithm to find the Strongly Connected Components (SCC). **04**

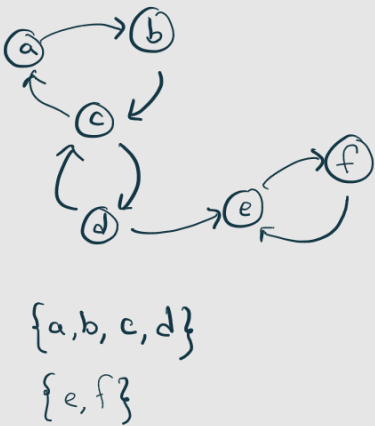
CO3 **ii.** Suppose a bridge gets **blocked**. **Explain** how you can efficiently calculate the updated SCCs from b(i) without rerunning the algorithm on the entire graph. **02**

CO3 c) A building is called "**Accessible**" if it's possible to start from there, **visit some (or all)** of the other buildings, and eventually return to that building. **03**

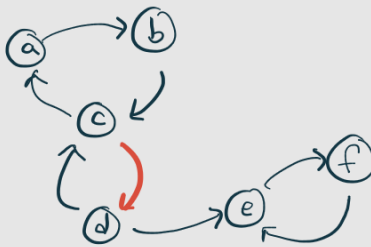
Propose an algorithm with a pseudocode/programmable code/step-by-step logical instructions to determine whether every building on the campus is **Accessible**. The algorithm should run in **at most** $O(V \times (V + E))$ time.

Row column বসান
arr[r][c]
direct value আছে
O(1)

b(ii) **ব্রীজ ব্লক** SCC না করে যে Bridge blocked শুরু ওইটার SCC করবে। Example:



যদি একটি Bridge block হয়ে যায়



{a, b, c, d}

→ ওই SCC এর Bridge blocked তাই শুরু ওই SCC change হবে

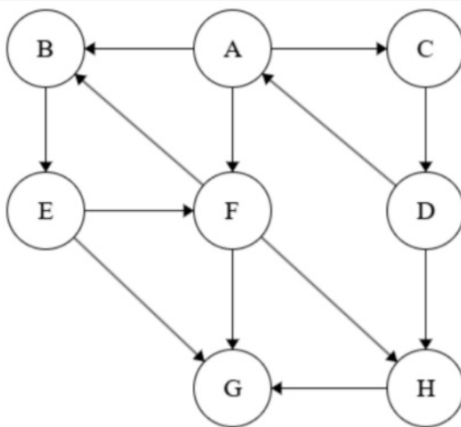


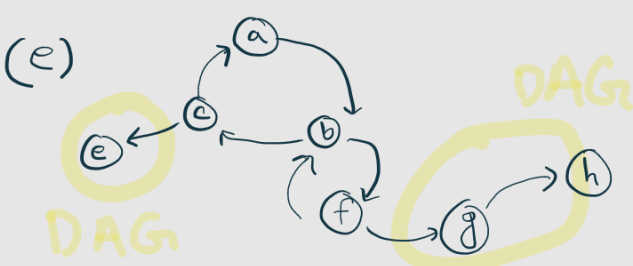
Figure 1: Q1(b)

c. **Simulate** Kosaraju's algorithm on the given graph Q1(b) to find the strongly connected components (SCCs). Label each SCC with the set of nodes it contains. While performing graph traversal, use node A as the starting node and break ties in alphabetical order. **04**

d. Now, treat each SCC identified in Q1(c) as a single "**super node**." In this new graph, every edge from the original graph either becomes internal (and thus disappears, since it stays within an SCC) or becomes an edge connecting two different SCCs. This transformation results in a Directed Acyclic Graph (DAG). **Draw** this DAG. **01**

e. Given a set of SCCs, what is the maximum number of **new** SCCs that can appear if we **delete one edge** from the DAG of these SCCs? **01**

→ আগের page এর last question. same answer



DAG থেকে delete করলে কোন new Edge হবে না। কারণ DAG পাল্টে এ cycle নাই

1. Dhaka, especially your area, is facing an Electricity crisis. Many initiatives including load shedding, reducing office hours, etc have already been taken. The local authority of your area is looking for feasible solutions to reduce the consumption of electricity. One of the members of their advisory committee has suggested that they should reduce the operating costs of road lighting. Till now every road is illuminated all night long. To reduce electricity consumption, they have decided to no longer illuminate every road, but to switch off the road lighting of some roads. To make sure that the inhabitants of your area still feel safe, they want to optimize the lighting in such a way that after darkening some roads at night, there will still be at least one illuminated path from every major point in your area to every other major point.

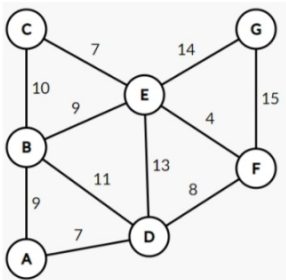


Figure : Graph of your area where the Vertices denote major points and Edges denote electricity consumption in a day.

- CO3

a.

Apply a suitable Algorithm to help the authority determine the roads they need to illuminate and the cost which will minimize the electricity consumption. If you need to consider any root vertex for your Algorithm, you can consider 'A' as such.

6
- CO5

b.

Observe the Graph of your area and your provided solution of question 1(a) and determine the maximum electricity of a day the authority can save.

1
- CO6

c.

The United Arab Emirates (UAE), especially Dubai, is hosting the Asia Cup 2022. They too want to minimize the electricity consumption by switching off the lights in some roads leaving exactly one illuminated path from every major point in your area to every other major point. However, to keep the tourists entertained, the Authority of Dubai has taken an opposite approach to yours. They want to illuminate the roads which require the most electricity.

3
- Propose an algorithm to compute the path which fulfills this requirement.

MST

opposite approach

w	(u,v)
15	(F,G)

বড় গ্রাফ শুরু

- CO3

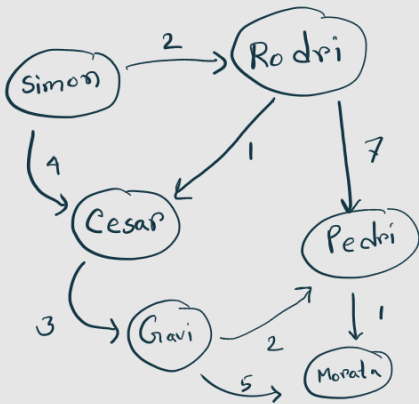
a.

As we all know, the Spanish National Football Team is notorious for their passing football, especially their back passes. After their disastrous 2022 World Cup campaign, The King of Spain made you the coach and declared that no back passes are allowed on the field from now on till Spain scores 2 goals, that is, Simon the goalkeeper can only pass to the defenders Rodri and Cesar, but Rodri and Cesar cannot pass the ball back to Simon. Also, every pass now comes with a cost– and the more cost players accumulate, the angrier the King gets. These are the costs of the passes:

06
- [Fall 22]

Passes	Cost
Simon → Rodri	2
Simon → Cesar	4
Rodri → Cesar	1
Rodri → Pedri	7
Cesar → Gavi	3
Gavi → Pedri	2
Pedri → Morata	1
Gavi → Morata	5

Find the lowest cost for the ball to reach every player starting from Simon, using any algorithm you find useful. The King will need to see every single step of the algorithm, or he will not believe you and fire you on the spot.



MST (Every player - Means সব node এ যতে হবে। আর MST সব Node এ যায়

(b)

Passes
Simon → Rodri
Simon → Cesar
Rodri → Busquets
Rodri → Pedri
Cesar → Rodri
Busquets → Gavi
Busquets → Pedri
Gavi → Cesar
Gavi → Pedri
Gavi → Fati
Fati → Morata
Pedri → Fati
Morata → Pedri

Now using a suitable algorithm, find out the largest group of players who can pass the ball among themselves. For example, one such group can be (Fati, Morata, Pedri) where Morata can pass to Pedri, Pedri can pass to Fati and Fati can pass to Morata. It is important to keep in mind that the king will be observing every step of the algorithm. Any discrepancies can lead you to lose your job.

Group বলছে SCC

Graph এর first এ আঁকবা